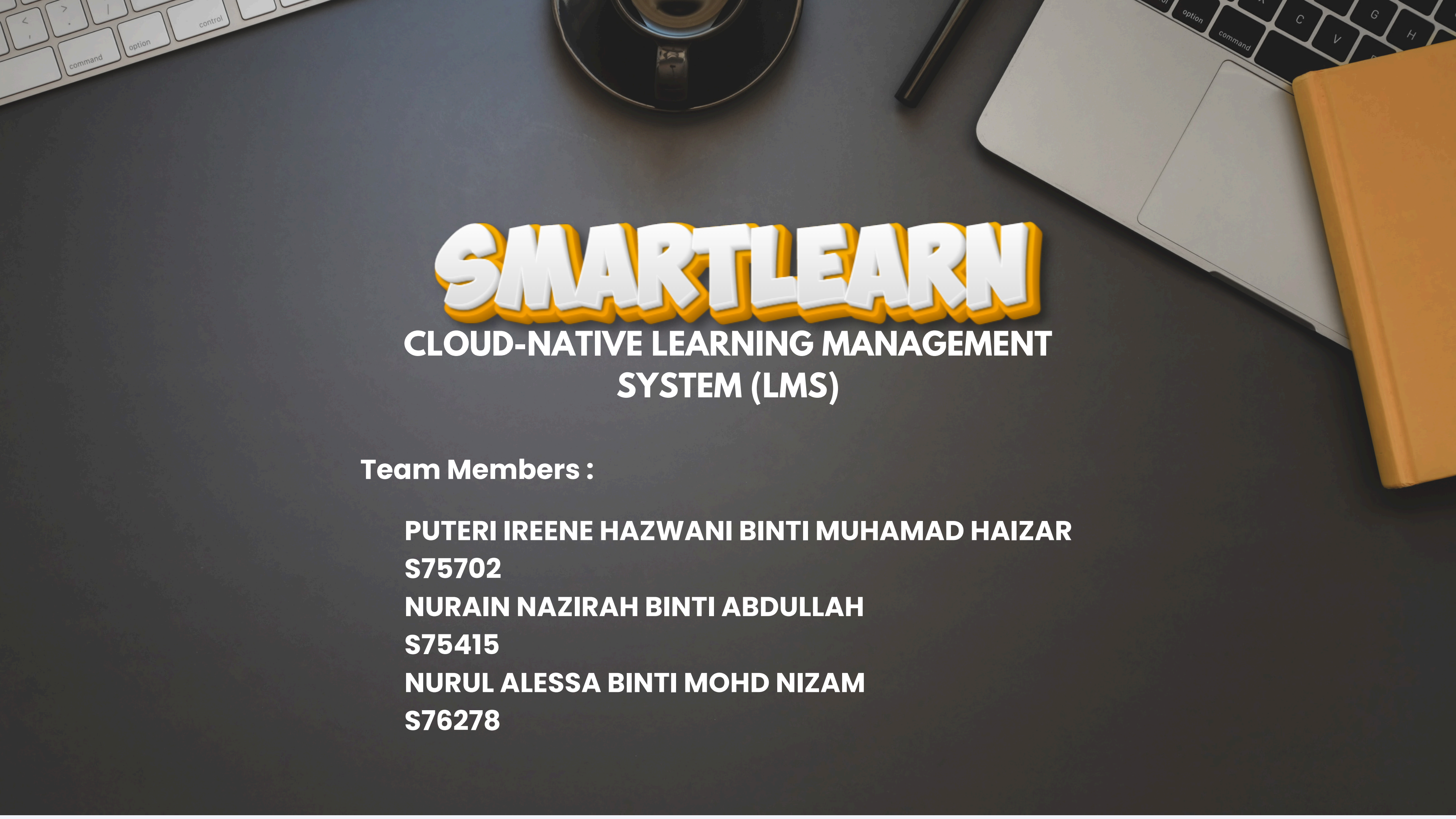




SMARTLEARN

CLOUD-NATIVE LEARNING MANAGEMENT SYSTEM (LMS)

Team Members :

PUTERI IREENE HAZWANI BINTI MUHAMAD HAIZAR

S75702

NURAIN NAZIRAH BINTI ABDULLAH

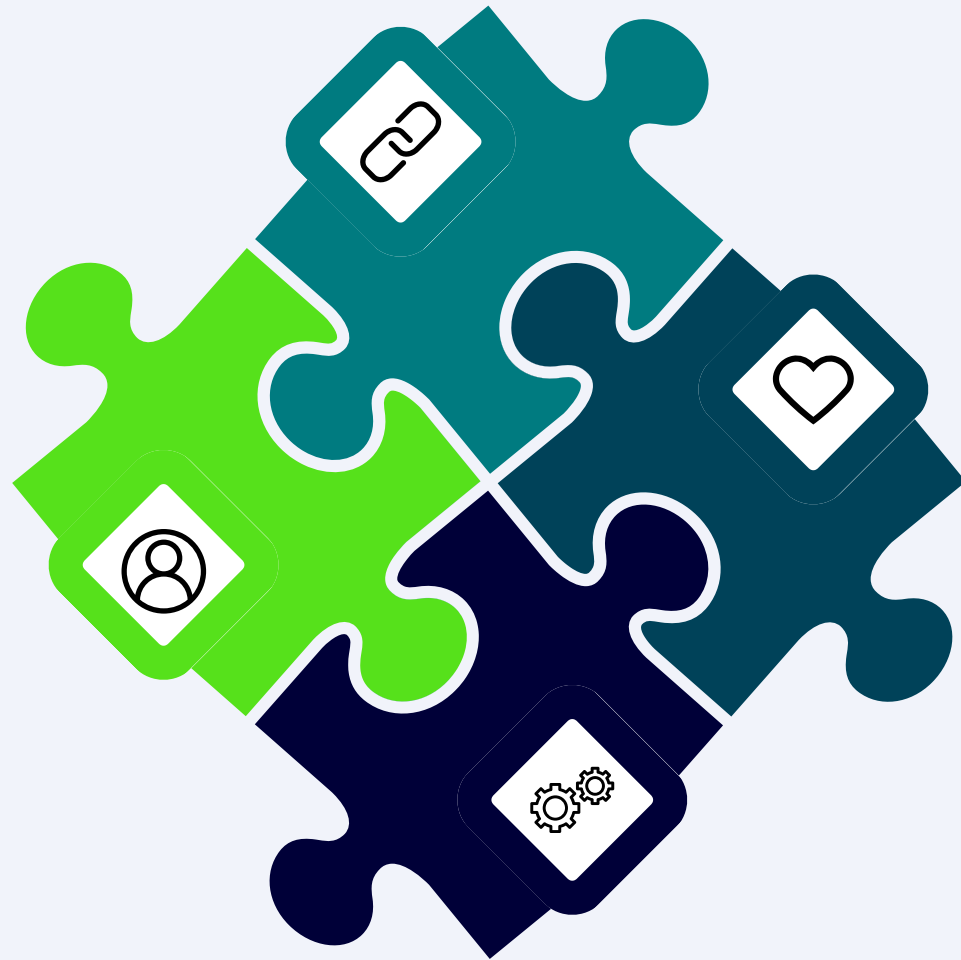
S75415

NURUL ALESSA BINTI MOHD NIZAM

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Introduction

SMARTLEARN is a modern cloud-native Learning Management System designed to support online teaching and learning activities. The system enables students to register, join courses, submit assignments, and view grades, while lecturers can manage courses, create assignments, and evaluate student performance.



Problem Statement

- Traditional LMS often experience slow performance during peak usage.
- Monolithic architecture makes maintenance and updates difficult.
- Limited scalability affects system growth and user experience.

Project Objective

Develop a cloud-native LMS that is:

- Scalable
- Maintainable
- Secure
- User-friendly

Key Features

- User Registration & Login
- Course Enrollment
- Assignment Submission
- Grading Management

3-Tier Architecture

01

Presentation Layer

- Web Application (Single Page Application)

02

Application Layer

- User Service
- Course Service
- Assignment Service
- Grading Service
- Security Component
- Notification Component

03

Data Layer

- User Database
- Course Database
- Assignment Database
- Grade Database

04

Benefits

- Separation of Concerns
- Better Maintainability
- Improved Security
- Easier Scalability

Cloud Deployment Architecture



Frontend

01

- Vercel



Database

02

MongoDB



Notification

03

- Amazon Simple Mail Services (AWS SES)



Backend

04

- Node.js
- Express.js
- Render
- React.js



Security

05

- JSON Web Token (JWT) Authentication



Advantages

06

- High Availability
- Better Performance
- Cloud Scalability

MEMBER RESPONSIBILITY ON PROJECT

Team Member	Role	Responsibilities
Puteri Ireene Hazwani binti Muhamad Haizar	Frontend Developer	• Design responsive user interfaces (UI). • Build client views (login gates, dashboards, layouts). • Configure local VS Code environment setup. • Manage client API fetches and localStorage tokens.
Nurain Nazirah binti Abdullah	Backend Developer	• Construct server-side routes. • Handle data request transformations. • Process backend payload deliveries. • Manage hosting pipelines on Render.
Nurul Alessa binti Mohd Nizam	Full-stack Architect & Database Lead	• Orchestrate the 3-tier architecture framework. • Connect and manage database structures. • Configure cross-domain CORS rules. • Validate cloud communication pathways.

LESSONS LEARNED

Session Security

Mastered saving JWT tokens in localStorage to keep users logged in and handle secure page redirects.

Cloud Deployment

Learned to configure environment variables and fix cross-origin network issues (CORS errors) on production servers.

Error Debugging

Realized how to use terminal shortcuts and AI tools to quickly isolate and fix broken runtime code.

API Integration

Gained experience connecting a frontend client (Vercel) to a backend server (Render) using fetch requests.

Responsive Design

Learned to use CSS Flexbox and Grids to stop UI layouts and tables from breaking on mobile screens.

